Week 6: Practical session on atomistic simulation using LAMMPS and OVITO

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1 LAMMPS and OVITO

LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator) is a classical molecular dynamics code with a focus on materials modeling for solid-state materials (metals, semiconductors) and soft matter (biomolecules, polymers) and coarse-grained or mesoscopic systems. It can be used to model atoms or, more generically, as a parallel particle simulator at the atomic, meso, or continuum scale. LAMMPS runs on single processors or in parallel using message-passing techniques and a spatial-decomposition of the simulation domain. Many of its models have versions that provide accelerated performance on CPUs, GPUs, and Intel Xeon Phis. The code is designed to be easy to modify or extend with new functionality. Further information can be found at www.lammps.org

To run LAMMPS you need to be in a directory containing an lammps input file `in_*.lammps` and also file defining the empirical interatomic force. In both Windows and Linux you will need to be in terminal session. For Windows this referred to as the "Command shell" or "PowerShell" which is accessed via the `windows key + X`. When in this terminal session, the command `dir` will list the contents of the directory you are in, and the command `cd newDir` will select a new directory to be `newDir`. To run LAMMPS you would type:

```
lmp -in in_example.lammps
```

where `in_example.lammps` is the LAMMPS script. For Linux/Mac machines one might need to be replace the above with the full path. All information about LAMMPS and how to run it may be found at docs.lammps.org

OVITO is a scientific visualization and analysis software for molecular and other particle-based dataset, typically generated by numeric simulation models in materials science, physics and chemistry disciplines. The online user manual may be found here

This tutorial script assumes you have LAMMPS and OVITO-lite installed on your computer

2 Exercise 1: FCC Al NPT equilibriation and anneal

Go to the directory `exercise1`.

In this exercise you will be setting up and equilibriating a perfect Al FCC crystal containing 2000 atoms using an NPT ensemble (a simulation at fixed particle number, fized (zero) pressure and fixed temperature). The LAMMPS input script is `in_fccAl_anneal.lammps` and the empirical interatomic potential for FCC Al is given in the file `Al99.eam.alloy`.

As a start, you should read through `in_fccAl_anneal.lammps` to fully understand the nature of the simulation. The basic structure of the LAMMPS contains the following aspects:

1. Set up the simulation by defining boundary conditions, the used units, the time-step, the Al interatomic potential and the FCC lattice.

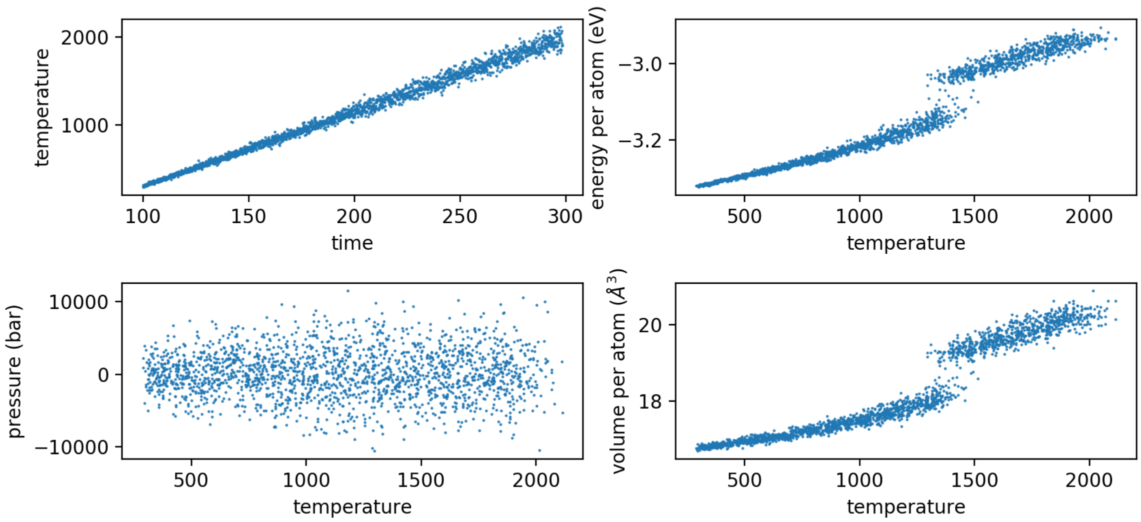

Figure 1: Plot of (left upper panel) temperature versus time, (right upper panel) energy per atom versus temperature, (left lower panel) pressure versus temperature, and (right lower panel) volume per atom versus temperature.

2. Equilibriation of the simulation cell to 300K at zero pressure. Thermodynamic output is stored in `log_equil.lammps`. This part of the simulation is done for 10000*`timestep`=10 psecs
3. The annealing of the simulation cell to 2000K. Thermodynamic output is stored in `log_anneal.lammps`. This part of the simulation is done for 20000*`timestep`=20 psecs

The equilibriation is performed by `fix 1 all npt temp 300.0 300.0 0.001 iso 0.0 0.0 0.1` which sets the start and end temperature to 300K and the start and end pressure to zero. The annealing to 2000K is performed by the similar command `fix 1 all npt temp 300.0 2000.0 0.001 iso 0.0 0.0 0.1`

There exists a python script in the same directory which will generate plots associated with the annealing part of the simulations — see fig. 1. Describe what you see.

To visualize the atomic scale processes underlying the thermodynamic data of fig. 1, load the dump file containing every 100th configuration of the anneal `dump_300-2000K_npt.atom` into OVITO. You should have 200 configurations loaded in, which can be cycled through by clicking on the "play" button located in the lower middle region of the OVITO region. By watching the animation, it should be clear what is happening. To make it even clearer, one can add a OVITO modifier to color the atoms according to their local crystalline structure. Go to the "add modification" tab and scroll down to select "Ackland-Jones analysis". Make sure "Color atoms by type" is ticked. The atoms should now be colored according to their local crystal structure. At the lowest temperature most should be green indicating they are FCC. What happens at higher temperatures? Another useful modifier is "Coordination analysis". Select this and choose 8 Å for "Cutoff radius" and about 100 for the "Number of historgram bins". What is the difference between the low temperature and high temperature pair distribution function?

Note that the order of the OVITO modifiers is important.

Finally, add the modifier "Manual selection" and click on "pick". Now click once on a particle. It should change it's color to red. Now, add the modifier "Generate trajectories lines" and choose "selected particle", and click on "Generate trajectory lines". You should see the trajectory of the selected particle during the whole anneal. How does its mobility change as a function of temperature?

How would the curves change in fig. 1 if you ran the anneal of 200000 steps rather than 20000 steps?

3 Exercise 2: Interaction between two vacancies in FCC Al

Go to the directory `exercise2`.

In the lecture on MD, you already encountered the simulation of a single vacancy — by removing one atom from an otherwise perfect lattice, non-zero forces exist on the surrounding atom, which when relaxed produce a slightly modified structure with all atoms having a zero-force. This is the vacancy defect. If one starts with a perfect FCC lattice containing N atoms, and then removes one atom, the energy of such a vacancy defect is defined as

$$E_{\mathrm{vacancy}} - E_{\mathrm{fcc}} \times (N-1), \tag{1}$$

where E_{vacancy} is the total potential energy of the simulation cell after relaxation, and E_{fcc} is the total energy per atom of the perfect FCC lattice. The LAMMPS input script `in_fccAl_vacancy.lammps` will calculate this value and output it in the appropriate log file.

The LAMMPS script starts with a perfect lattice, calculates its total energy, removes an atom, relaxes the structure to a new zero-force configuration, and evaluates eqn. 1 for both the un-relaxed and relaxed structure. Both these configurations are saved for visualization in OVITO. To visualize the (very small) strain around the vacancy, load `final_state.lammps` into OVITO, and select the modifier "Elastic strain modifier" and select the reference lattice as being FCC with a lattice constant of 4.05 Angstroms. Now select the modifier, "Color coding" and choose as "Input property" volumetric strain. Click on "Adjust range" to make sure you have a good colouring scheme. To see the vacancy and its strain field you need to remove some of the atoms. To do this, select the "Expression select" modifier, and enter `abs(VolumetricStrain)<1e-4` as the "Boolean expression". The atoms satisfying this condition should be colored red. To remove them, select the "Delete selected" modifier. Now you can see the local volumetric strain around the vacancy.

Note that the LAMMPS script calculates the unrelaxed and relaxed vacancy defect energy, showing that the creation of strain lowers the energy by about 0.05 eV.

Such strain fields can lead to an interaction between defects, and here we consider such an interaction between two vacancies. To create two vacancies in your FCC simulation cell, you can add a second particle-to-be-removed via the modified LAMMPS command `group vacancy id 15239 15004` in the input script `in_fccAl_vacancy.lammps`. Here the two particles with id's 15239 and 15004 are removed. By doing this, the LAMMPS script will calculate the defect energy of a di-vacancy. If the vacancies are sufficiently far apart, then this energy should be twice that of the single vacancy defect energy. However, as they near each other, this should change. The LAMMPS script will now calculate and output the di-vacancy defect energy defined as:

$$E_{\mathrm{di-vacancy}} - E_{\mathrm{fcc}} \times (N-2). \tag{2}$$

By choosing the second atom via OVITO, investigate how the defect energy of the di-vacancy changes and establish if the di-vacancy is stable defect — do the two vacancies bind to form a stable complex? To find the appropriate atom id using OVITO, select the arrow-cursor mode (see upper left panel of buttons in OVITO) and move your cursor over an atom. Doing so, will reveal the atom information on the lower part of the screen.

4 Exercise 3: Interstitial diffusion in BCC Fe and BCC W

Go to the directory `exercise3`.

To directly simulate the diffusion of vacancies using MD is not possible due to the time-scale problem. One type of defect that diffuses sufficiently fast to be seen in at the nano-second time-scale is that of an interstitial. An interstitial defect is an additional atom placed within an otherwise perfect lattice — see fig. 2. Unlike the vacancy, there is more than one possible structure for the interstitial defect to take with the lowest energy structure being the most important. Which structure this is, will depend on the symmetry of the lattice and also on the nature of the interatomic force.

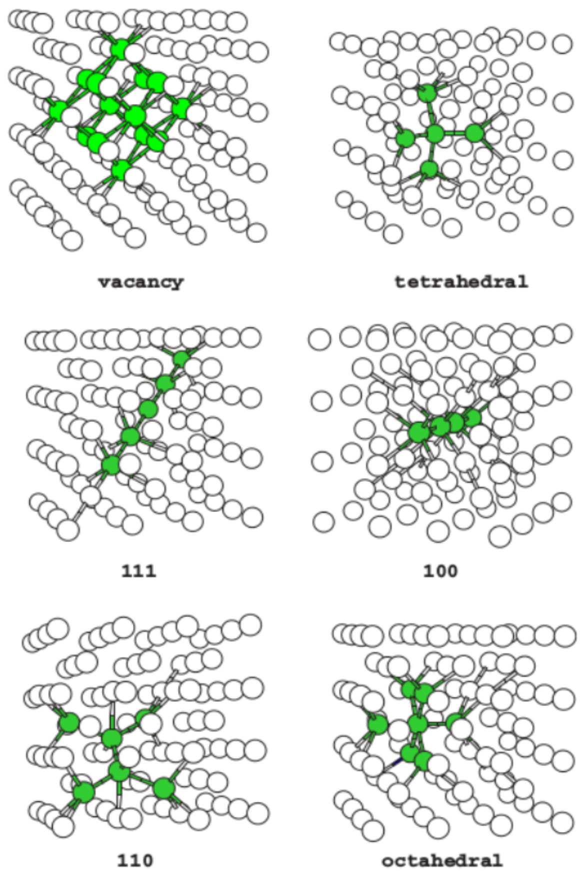

Figure 2: Vacancy and possible interstitial structures for the BCC lattice. Those atoms highlighted in green are atoms whose local energy is much larger than that of the perfect BCC lattice.

Somewhat similar to the vacancy defect, the energy of the interstitial defect is given by

$$E_{\text{interstitial}} - E_{\text{bcc}} \times (N + 1) \tag{3}$$

Two LAMMPS simulations involving one interstitial will be performed, one for BCC Tungsten and one for BCC Fe. For BCC W, the lowest energy interstitial structure is the so-called 111 (crowdian) interstitial defect, whereas for BCC Fe it is the 110 (dumbel) interstitial, see fig. 2. The correspinding LAMMPS input scripts are `in_W_interstitial.lammps` and `in_Fe_interstitial.lammps`. These are in the corresponding directories `W111` and `Fe110`. To create the interstitial atom, an additional atom is added to the otherwise perfect BCC lattice via `create_atoms 1 single 0.7913 0.7913 0.7913` for W and `create_atoms 1 single 1.4332 1.4332 0.0` for Fe. The last three numbers represent the absolute (x, y, z) coordinates where the additional atom is inserted. The differences between W and Fe reflect the 111 and 110 chains of atoms into which the extra atom is inserted.

The two LAMMPS scripts relax the initial structure to a new zero-force structure. Use OVITO to visualize the resulting strain field in a similar way as you had done for the vacancy.

To visualize the strain around the interstial, load `final_state.lammps` into OVITO, and select the modifier “Elastic strain modifier” and select the reference lattice as being BCC with a lattice constant of 3.1651956 Angstroms for W and 2.8665 Angstroms for Fe. Now select the modifier, “Color coding” and choose as “Input property” volumetric strain. Click on “Adjust range” to make sure you have a good colouring scheme. To see the vacancy and its strain field you need to remove some of the atoms.

To do this, select the "Expression select" modifier, and enter `abs(VolumetricStrain)<0.0001` as the "Boolean expression". To remove them, select the "Delete selected" modifier. Now you can see the local volumetric strain around the interstitial.

What is different when compared to the vacancy? How does the interstitial defect energy compare to the vacancy? Relaxed and unrelaxed? What do you think is the reason for their differences. Both scripts then go on to equilibriate the sample at 700K for Fe and 1500K for W, and then perform 20 picoseconds of simulation, storing the atomic configurations every 0.1 picoseconds in the associated dump files. To better see the diffusing interstitial go to the "add modification" tab and scroll down to select "Common neighbour analysis". Make sure "Color atoms by type" is ticked and select the option "Adaptive CNA". Now add the modifier "Select type". Make sure the property is "Structure" and tick the BCC type. All BCC atoms should now be red. To remove them, select the "Delete selected" modifier. You now should be able to see the core region of the interstitial, and how it diffuses through the lattice.

What is the difference between Fe and W? Which one is the faster?

5 Exercise 4: Creating a model CuZr metallic glass

A model metallic glass structure is produced through atomistic simulation in a similar way as done in experiment — an under-cooled liquid is equilibriated and rapidly quenched. As the temperature drops, the atoms of the under-cooled liquid slow down, eventually freezing into an amorphous configuration which changes little at the time-scale of the quench. The temperature at which this occurs is referred to as the glass temperature (regime).

The LAMMPS script `in_CuZr_quench.lammps` performs this quench for a model Cu-Zr binary alloy system with equal populations of each chemical element. The script has the following structure:

1. An $L1_0$ binary structure is created and equilibriated at 4000K under NVT conditions for 10 pico-seconds. What do you think happens to this initial structure? Is 10 pico-seconds long enough to equilibriate this high temperature regime?

2. The temperature is reduced to 2000K under NVT conditions and further equilibriated for 10 pico-seconds.

3. The quench is performed by linearly reducing the temperature from 2000K down to 10K in 100 pico-seconds under zero-pressure NPT conditions. This similation might take some time. During this part of the simulation the configuration is saved every 1 pico-second in the dump file `dump_2000-10K_npt.atom` for visualization in OVITO

With the above, you now have produced a CuZr binary glass. To look at the thermodynamic variable evolution use `plot_lammps.py`. From the volume and energy data as a function of temperature, can you determine the temperature regime of the glass transition.

Load into OVITO the dump file `dump_2000-10K_npt.atom` and watch the glass formation dynamics. Use the trajectory modifier to investigate the path of one particle. What happens when the glass is formed? Compare the temperature at which this happens to the glass transition temperature derived from the thermodynamic data.

The file `log_quench_1nanosec.lammps` contains the thermodynamic data for a slower quench occurring over 1 nano-second. Use the supplied python plotting script to compare this slower quench data. What is different and why?

Using the 100 pico-second data you generated yourself and the supplied 1 nano-second quench data `dump_2000-10K_npt_1nanosec.atom` with OVITO, apply some of the structural analysis modifiers to the structural evolution ("Ackland-Jones analysis" and/or "Common neighbour analysis"). What local structural motives appear at and below the glass transition temperature regime? Is there a difference between the 100 pico-second and 1 nano-second structures?

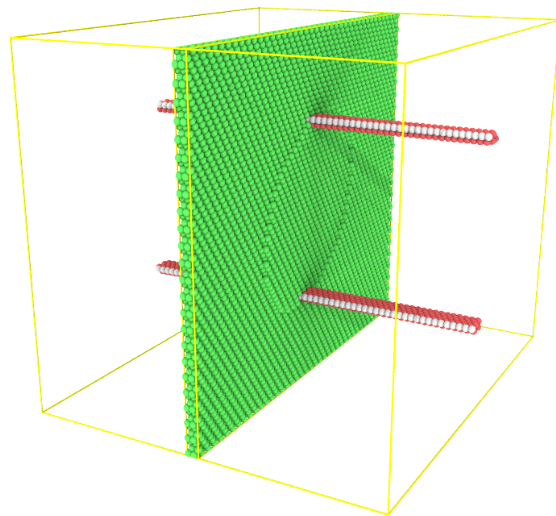

Figure 3: Starting structure consisting of two screw dislocations. The green region is the actual simulation cell.

An interesting comparison is to quench pure Cu. This is done most easily by starting with a pure Cu $L1_0$ structure via: `create_atoms 2 region box basis 1 1 basis 2 1 basis 3 1 basis 4 1`. Run the simulation and look at the thermodynamic data. Also included in this directory is the dump file for a 1 nanosecond quench for pure Cu. What do you expect to happen? How does this data differ from the binary alloy data? Look at the pure Cu dump file in OVITO. What happens?

6 Exercise 5: Mobility of screw dislocations in FCC Cu

In this last simulation you will investigate the mobility of a screw dislocation in FCC Cu. Because the simulation will be fully periodic, two screw dislocations (with opposite Burger's vector directions) will in fact be simulated. The python program `createDislocationDipole.py` will create a configuration that can be read in by LAMMPS and OVITO, using the simple theory covered in the lectures. Specifically, the displacement field given by

$$u_x = \frac{b}{2\pi} \tan^{-1}\left[\frac{y}{z}\right] \text{ and } u_y = u_z = 0, \tag{4}$$

will be applied to a perfect FCC lattice whose cartesian axes are along the $[1\bar{1}0]$, $[112]$ and $[11\bar{1}]$ crystallographic directions. The LAMMPS script `in_dislocationDipole.lammps` reads this file in, finds a new zero-force configuration and then performs molecular dynamics at 300K using NVT for 100 pico-seconds. Configurations stored every 0.1 pico-seconds are saved in the dump file `dislocationDipole.atom`. I suggest you change the random seed in the command `velocity all create 300.0 12345`, since then you will have different results from your colleagues.

Fig. 3 shows the initial relaxed structure coloring the atoms according to their "Common neighbour analysis". The actual simulation cell is quite short along the direction of the [110] Burger's vector (the line direction of the screw dislocation). Periodic images of only the dislocation core are shown using the OVITO "Replicate" modifier along the x-axis.

Load in the dump file and find the best way to visualize the two dislocations. What happens as a function of time? What well known screw dislocation process brings the two dislocations together to eventually annihilate? Does you simulation show annihilation. How do you think this will differ for a true three dimensional simulation? Would annihilation occur more quickly in 3D?